Ali Suvizi

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OVERVIEW

Ph.D. researcher specializing in computer architecture, hardware–software co-design, FPGA/SoC systems, chiplets, and heterogeneous accelerators. Experienced in FPGA prototyping, embedded firmware, C/C++ architecture simulation, and performance optimization. First-author publications at ACM ICS and GLSVLSI, with cross-institutional research collaborations.

EDUCATION

George Washington University, Washington, DC Aug 2023 – Dec 2027 (expected)
Ph.D. in Computer Engineering | *Advisor: Prof. Guru Prasadh Venkataramani*
Research Interests: High-Performance Computing, Computer Architecture, Machine Learning, Digital Design, LLMs

Amirkabir University of Technology, Tehran, Iran 2019 – 2022
M.Sc. in Computer Architecture

NOTABLE PROJECTS

Swift-Healer: Firmware-Reconfigurable Self-Healing (ACM GLSVLSI 2026)
Implemented FPGA/SoC glitch resilience with GRU prediction (**F1 = 0.87**) and **5 μs–5 ms** recovery.
Languages: C/C++, Python | **Tools:** Vivado/Vitis HLS, AXI DMA | **Platform:** Zynq-7000
Link: Published Paper

Auto-Healer: Fault-Tolerant CNN Acceleration (ACM ICS 2025)
Implemented FPGA CNN recovery using DMR/checkpointing, achieving **40 ns** transient and **120 ns** permanent repair.
Language: VHDL | **Methods:** RTL, DMR, LeNet-5 | **Platform:** FPGA
Link: Published Paper

PIM and FPGA: Low-Latency Edge AI (IEEE Cloud Summit 2024)
Benchmarked LeNet-5 on PIM (**92**× CPU speedup) and FPGA (**0.02 ms** inference latency).
Language: VHDL | **Tools:** MNSIM, RTL | **Platforms:** FPGA, simulated PIM
Link: Published Paper

LPCR: Low-Power Network-on-Chip Links (IEEE TCAD 2025)
Reduced serial NoC switching by up to **13.2%** and link energy by up to **9.9%** through code reordering.
Methods: Information Theory, Huffman Coding, Low-Power NoC | **Platform:** Serial NoC Links
Link: Published Paper

FD-Hybrid: Chiplet-Aware Scheduling (IEEE Cloud Summit 2026)
Co-developed AoI-aware chiplet scheduling for freshness, latency, and deadlines; **Best Paper Award**.
Methods: AoI, Task Scheduling, Performance Modeling | **Platform:** Chiplet-Based Systems

LLMs on a Budget: Efficient Fine-Tuning (IEEE OJCS 2025)
Evaluated LoRA/BF16 and GPU power capping: **over 40%** memory savings; **5.55%** throughput loss at 300 W.
Language: Python | **Framework:** PyTorch, LoRA | **Platform:** NVIDIA RTX 4090
Link: Published Paper

MIPS Processor: RTL Design and ASIC Synthesis
Designed Verilog MIPS RTL and optimized ASIC synthesis using Synopsys Design Compiler and Cadence tools.
Language: Verilog | **Tools:** Design Compiler, SoC Encounter | **Platform:** ASIC
Link: GitHub Repository

CNN Implementation, Quantization, and Hardware Security
Implemented FPGA CNNs and PUF-secured LeNet-5 in VHDL, plus quantized CPU/GPU CNNs in Python.
Languages: VHDL, Python | **Methods:** CNN, Quantization, PUF | **Platforms:** FPGA, CPU, GPU

WORKING EXPERIENCE

Graduate Research Assistant, George Washington University Aug 2023 – Present
Computer Architecture and Reliable Hardware Systems | Washington, DC

- **Computer Architecture and Performance Optimization:** Designed and evaluated chiplet-based and heterogeneous architectures using C/C++ simulation, analyzing performance, latency, power consumption, and energy efficiency.
- **FPGA/SoC and Hardware–Software Co-Design:** Developed FPGA accelerators integrating RTL design, ARM embedded firmware, AXI interfaces, and partial reconfiguration on Xilinx Zynq SoCs.
- **Hardware Reliability and Fault Tolerance:** Developed fault prediction, runtime detection, hardware redundancy, and self-healing recovery mechanisms for safety-critical computing systems.

PUBLICATIONS

- **A. Suvizi**, J. Iwu, K. Amberiadis, and G. Venkataramani. *Swift-Healer: Firmware-Reconfigurable Self-Healing for Remote Glitch-Injection on Autonomous Navigation Systems*. ACM GLSVLSI, 2026.
- E. Dogan, **A. Suvizi**, E. Erbayat, S. Subramaniam, and G. Venkataramani. *Exploring Information-Freshness v. Task-Latency Tradeoffs for Schedulers in Chiplet-based Hardware*. IEEE Cloud Summit, 2026. **Best Paper Award**.
- **A. Suvizi** and G. Venkataramani. *Auto-Healer: Self-Healing Hardware for Perception Stage Faults in Autonomous Driving Systems*. ACM ICS, 2025.
- M. Adelkhani, **A. Suvizi**, et al. *LPCR: Information-Theory-Based Low-Power Code Reordering for Serial Links in Network-on-Chip*. IEEE TCAD, 2025.
- K. Gogineni, **A. Suvizi**, and G. Venkataramani. *LLMs on a Budget: System-Level Approaches to Power-Efficient and Scalable Fine-Tuning*. IEEE Open Journal of the Computer Society, 2025.
- **A. Suvizi**, R. Ahmadi, M. Saheb Zamani, and M. R. Meybodi. *A Parallel Computational Approach for Energy-Efficient Hydraulic Analysis of Water Distribution Networks Using Learning Automata*. Sustainable Computing: Informatics and Systems, 2025.
- **A. Suvizi**, S. Subramaniam, T. Lan, and G. Venkataramani. *Exploring In-Memory Accelerators and FPGAs for Latency-Sensitive DNN Inference on Edge Servers*. IEEE Cloud Summit, 2024.
- **A. Suvizi**, A. Farghadan, and M. Saheb Zamani. *A Parallel Computing Architecture Based on Cellular Automata for Hydraulic Analysis of Water Distribution Networks*. Journal of Parallel and Distributed Computing, 2023.

Submitted manuscript (IEEE TDSC): A. A. Malik, **A. Suvizi**, G. Venkataramani, and A. Aysu. *Emulated Integrity Replica: Enabling Self-Healing on FPGA SoCs via Hierarchical Twins*. Submitted to *IEEE Transactions on Dependable and Secure Computing*; arXiv preprint, 2026.

TECHNICAL SKILLS

Programming & Systems: C, C++, Python, Java, Assembly, CUDA, OpenMP, MPI, embedded ARM firmware, RISC-V.

Frameworks: PyTorch, TensorFlow, GEM5, Intel Parallel Studio, Intel VTune Profiler, Intel Perf, NVIDIA Nsight, SST Simulator, HP BYOD, Docker.

Hardware Design & EDA: FPGA, SoC, VHDL, Verilog, AMD Xilinx Vivado, AMD Xilinx Vitis HLS, ModelSim, MATLAB, Synopsys Design Compiler, SPICE.

HONORS AND PROFESSIONAL SERVICE

Honors:

- **U.S. Permanent Residency (EB-2 National Interest Waiver)**, approved 2026.
- **Best Paper Award, IEEE Cloud Summit (2026)**
- **IEEE DAC** Conference Travel Grant (2026).
- **GWU** University Fellowship (2023, 2024, 2025, 2026)
- **gem5** Bootcamp Student Travel Grant (2024)
- Ranked 1st in the graduating M.Sc. class at Amirkabir University of Technology.

Peer reviewer: IEEE Micro; IEEE Computer Architecture Letters; IEEE Computer; IEEE Transactions on Sustainable Computing; IEEE Access; Journal of Hardware and Systems Security.